Arrays:

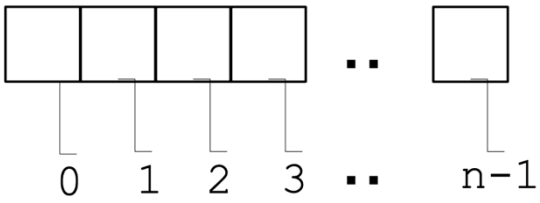
An array is an indexed list of values that all have the **same type**.

You can make an array of any type int, double, String, etc.

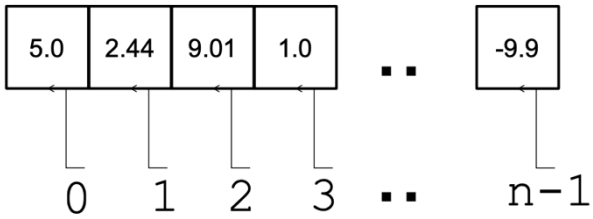
In array, the memory allocated is contiguous.

1. One-dimensional Array:

It is an array that has one index. **The index starts at zero and ends at length-1.**



Following an example of array of type double:



How to declare, initialize and access array?

An array is **defined** as following:

data_type array_name[];

OR

data_type[] array_name;

```
int[] values; // declare array of type int named “values”
int values[]; // same as above
```

To create an array of a given size, here size specifies the number of elements in the array, use the operator **new**:

```
double[] c = new double[5]; // declare an double array named “c” of size 5
double c[] = new double[5]; // same as above
```

Curly braces { } can be used to **initialize** an array. It can **ONLY** be used when you declare the variable.

```
int[] values = { 12, 24, -23, 47 }; // size of array deduced from the number of items
int values[] = { 12, 24, -23, 47 }; // same as above
```

Each array has a **length** variable built-in that contains the length of the array.

```
int[] values1 = new int[12]; // Declare and allocate a 12 elements int
array
int num = values1.length; // num is 12

int[] values2 = { 1, 2, 3, 4, 5 }
int size = values2.length; // size is 5
```

To access the elements of an array, use the [ ] operator. This operator contains the index value.

```
int[] values = { 12, 24, -23, 47 };
values[3] = 18; // {12, 24, -23, 18}
int x = values[1] + 3; // x = 27
```

Note: If the index is either outside the range of [0, length-1] or negative, Java Runtime will signal an exception called **ArrayIndexOutOfBoundException**.

Q> Is there an error in the following codes?

```
1. int[] values = {1, 2.5, 3, 3.5, 4};
2. char[] array = {'a', '*', '4', '/'};
   System.out.println(array[4]);
```

You can process all the elements of an array via a **loop.**

```
int[] values = { 12, 24, -23, 47 };
for (int i=0; i < values.length; i++)
{
System.out.println(values[i]);
}
```

```
12
24
-23
47
```

```
int[] values = { 12, 24, -23, 47 };
int i = 0;
while (i < values.length)
{
System.out.println(values[i]);
i++;
}
```

```
12
24
-23
47
```

```java
/* Java program to illustrate creating an array of type double and
   print each element with its index */
public class ArrayExample1 {
     public static void main(String[] args) {
     double[] array= {22.4, 13.3, -6.5, 9};
     for (int c = 0; c < array.length; c++)
     System.out.println("Element at index "+ c + " = " + array[c]);
     }
}
```

```
Problems | Javadoc | Declaration | Console | Progress
<terminated> ArrayExample1 [Java Application] /Library/Java/JavaVirtualMachin
Element at index 0 = 22.4
Element at index 1 = 13.3
Element at index 2 = -6.5
Element at index 3 = 9.0
```

Example: Write a Java program to find the summation of the elements in an array then print out the square root of that summation.

```java
public class ArrayExample2 {
     public static void main(String[] args) {
     int numbers[] = {8, 2, 6, 4, 3};
     int sum = 0;
     for (int i=0; i < numbers.length; i++)
          sum = sum + numbers[i];
     System.out.println("The summation is: " + sum);
     System.out.println("The square root is: " + Math.sqrt(sum));
     }
 }
```

```
Problems | Javadoc | Declaration | Console | Progress
<terminated> ArrayExample2 [Java Application] /Library/Java/JavaVirtualMachin
The summation is: 23
The square root is: 4.795831523312719
```

```java
// Java Example Read and Print Array
import java.util.Scanner;
public class ArrayExample3 {
   public static void main(String[] args) {

    // Declare variables
      int size;
      int[] items;
      Scanner in = new Scanner(System.in);

    // Enter the number of items i.e. the size of the array
      System.out.print("Enter the number of items: ");
      size = in.nextInt();

      items = new int[size]; // Allocate the array

      // Read the items of the array
      System.out.print("Enter the value of all items (separated
by space): ");
      for (int i = 0; i < items.length; i++)
         items[i] = in.nextInt();

      in.close();

    // Print array contents
      System.out.print("The array contains: ");
      for (int i = 0; i < items.length; i++)
       System.out.print(items[i] +" ");
   }
}
```

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```
<terminated> ArrayExample3 [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_112.jdk/Contents
Enter the number of items: 5
Enter the value of all items (separated by space): 33 8 25 -4 7
The array contains: 33 8 25 -4 7
```